Now we have three intervals. $(-\infty, 2), (2, 3)$ and $(3, \infty)$

(i) Take any point in $(-\infty, 2)$, say $x = 1$. Clearly $x^2 - 5x + 6$ is positive.
(ii) Take any point in $(2, 3)$, say $x = 2.5$. Clearly $x^2 - 5x + 6$ is negative.
(iii) Take any point in $(3, \infty)$, say $x = 4$. Clearly $x^2 - 5x + 6$ is positive.

For all x, in the intervals $(-\infty, 2)$ and $(3, \infty)$, $x^2 - 5x + 6$ is positive. At $x = 2, 3$ the value of $x^2 - 5x + 6$ is zero. Thus, $\sqrt{x^2 - 5x + 6}$ is defined for all x in $(-\infty, 2] \cup [3, \infty)$.
Hence the domain of $\sqrt{x^2 - 5x + 6}$ is $(-\infty, 2] \cup [3, \infty)$.

Example 1.22 Find the domain of $f(x) = \frac{1}{1-2\cos x}$.

Solution:
The function is defined for all $x \in \mathbb{R}$ except $1 - 2\cos x = 0$. That is, except $\cos x = \frac{1}{2}$. That is except $x = 2n\pi \pm \frac{\pi}{3}, n \in \mathbb{Z}$. Hence the domain is $\mathbb{R} - \{2n\pi \pm \frac{\pi}{3}\}, n \in \mathbb{Z}$

Example 1.23 Find the range of the function $f(x) = \frac{1}{1-3\cos x}$.

Solution:
Clearly,

$$\begin{array}{rrcccl} & -1 & \leq & \cos x & \leq & 1 \\ \Rightarrow & 3 & \geq & -3\cos x & \geq & -3 \\ \Rightarrow & -3 & \leq & -3\cos x & \leq & 3 \\ \Rightarrow & 1-3 & \leq & 1-3\cos x & \leq & 1+3 \end{array}$$

Thus we get $-2 \leq 1 - 3\cos x$ and $1 - 3\cos x \leq 4$.
By taking reciprocals, we get $\frac{1}{1-3\cos x} \leq -\frac{1}{2}$ and $\frac{1}{1-3\cos x} \geq \frac{1}{4}$.
Hence the range of f is $(-\infty, -\frac{1}{2}] \cup [\frac{1}{4}, \infty)$.

Example 1.24 Find the largest possible domain for the real valued function given by $f(x) = \frac{\sqrt{9-x^2}}{\sqrt{x^2-1}}$.

Solution:
If $x < -3$ or $x > 3$, then x^2 will be greater than 9 and hence $9 - x^2$ will become negative which has no square root in $\mathbb{R}$. So x must lie on the interval $[-3, 3]$.

Also if $x \geq -1$ and $x \leq 1$, then $x^2 - 1$ will become negative or zero. If it is negative, $x^2 - 1$ has no square root in $\mathbb{R}$. If it is zero, f is not defined. So x must lie outside $[-1, 1]$. That is, x must lie on $(-\infty, -1) \cup (1, \infty)$. Combining these two conditions, the largest possible domain for$f$ is $[-3, 3] \cap ((-\infty, -1) \cup (1, \infty))$. That is, $[-3, -1) \cup (1, 3]$.



Draw the number line and plot the intervals to get the required domain interval.

1.6.4 Operations on Functions

Composition

Let there be two functions f and g as given in the Figure 1.36 and Figure 1.37. Let us note that the co-domain of f and the domain of g are the same. Let us cut off Figure 1.37 of g and paste it on the Figure 1.36 of f so that the domain Y of g is pasted on co-domain Y of f. (See Figure 1.38.)